

# Human Impact Prediction – Dataset Requirements

Component 3: Machine-learning-based estimation of deaths, injuries, missing people, evacuated/displaced people, total affected people and human-impact risk. The table separates core model data from useful supporting GIS/environmental data.

| Factor                          | Dataset Needed                                                                                                                             | Where to Request in Sri Lanka                                                  | Format                                                              | Resolution                                                        | Purpose according to new component                                                                                                                                   |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Historical human impact         | Disaster event records: deaths, injured, missing, affected people, evacuated/displaced people, houses damaged; date, hazard type, location | Disaster Management Centre (DMC) / DesInventar                                 | CSV/Excel/database export; PDF reports if export unavailable        | Event + District/DS/GN level where available                      | Main training/target data for predicting human impact. Your component specifically needs historical deaths, injuries, missing, displacement and affected population. |
| Flood hazard                    | Flood occurrence, flood severity, probability/forecast output, affected area                                                               | DMC; Irrigation Department; Member 1's Flood Prediction component              | CSV/Excel; GeoJSON/Shapefile/GeoTIFF for spatial data               | Event/location dependent                                          | Provides flood severity and exposure information used as model inputs.                                                                                               |
| Landslide hazard                | Landslide occurrence, severity, probability/forecast output, affected area/hazard class                                                    | NBRO; Member 2's Landslide Prediction component                                | CSV/Excel; GeoTIFF/GeoJSON/Shapefile                                | NBRO maps commonly 1:50,000 and 1:10,000                          | Provides landslide severity and spatial hazard information for the human-impact model.                                                                               |
| Rainfall                        | Daily rainfall; preferably 24-hour, 3-day and 7-day accumulated rainfall; event rainfall                                                   | Department of Meteorology; NBRO; Irrigation Department hydrometric stations    | CSV/Excel; station time series                                      | Station-based; temporal resolution depends on station/data        | Captures hazard intensity/trigger conditions and helps explain variation in human impact.                                                                            |
| River water level               | River/gauge water level, alert/minor/major flood thresholds, rising/falling status                                                         | Irrigation Department                                                          | CSV/Excel/time series; PDF reports if historical export unavailable | Gauge/station; hourly or reporting interval where available       | Useful flood-intensity feature, especially for flood events.                                                                                                         |
| Population                      | Total population by GN/DS/District                                                                                                         | Department of Census and Statistics (DCS) – Census of Population and Housing   | Excel/CSV                                                           | GN level available in 2024 census                                 | Measures population exposure in the affected area.                                                                                                                   |
| Population age/vulnerability    | Children, elderly, sex/age groups and other available vulnerable groups                                                                    | DCS – Census of Population and Housing                                         | Excel/CSV                                                           | GN/DS/District depending on table                                 | Represents population vulnerability; helps distinguish areas with different potential human consequences.                                                            |
| Households / housing            | Number of households and housing units; optionally housing structure/type                                                                  | DCS – Census of Population and Housing                                         | Excel/CSV                                                           | GN/DS/District depending on table                                 | Supports exposure estimation and can improve prediction of affected/displaced people.                                                                                |
| Administrative boundaries       | Province, District, DS Division and GN Division polygons/codes                                                                             | Sri Lanka NSDI / Survey Department                                             | Shapefile/GeoJSON/GeoPackage/REST feature layer                     | National to GN level; NSDI services include GN and DS layers      | Links disaster records and population data to a common geographic unit.                                                                                              |
| Elevation                       | Digital Elevation Model (DEM)                                                                                                              | Survey Department/NSDI where available; otherwise an accepted DEM such as SRTM | GeoTIFF                                                             | Use the finest consistent resolution available for the study area | Geographic factor for exposure and terrain-related vulnerability.                                                                                                    |
| Slope                           | Slope derived from DEM                                                                                                                     | Derived by your research team from DEM                                         | GeoTIFF/raster                                                      | Same base resolution as DEM                                       | Represents terrain characteristics relevant to landslide exposure and geographic vulnerability.                                                                      |
| Distance from river             | Distance from rivers/streams                                                                                                               | NSDI/Survey Department river layers; derived in GIS                            | Shapefile/GeoJSON + derived raster/table                            | Vector/raster; derived at study resolution                        | Helps represent flood exposure of settlements/populations.                                                                                                           |
| Flood extent                    | Historical inundation/flood-affected polygons or raster                                                                                    | DMC / Irrigation Department / relevant flood studies                           | GeoTIFF/GeoJSON/Shapefile                                           | Event/map dependent                                               | Estimates which population areas were physically exposed to flooding.                                                                                                |
| Landslide hazard/susceptibility | NBRO landslide hazard zonation map/class                                                                                                   | National Building Research Organisation (NBRO)                                 | GeoTIFF/GeoPDF/Map/GeoJSON where available                          | NBRO publishes 1:50,000 and 1:10,000 mapping                      | Adds spatial landslide susceptibility/hazard context to human-impact prediction.                                                                                     |
| Historical disaster frequency   | Number of previous flood/landslide events for location and/or recent-period frequency                                                      | Derived from DMC/DesInventar historical records                                | CSV/Excel                                                           | GN/DS/District/event level depending on source                    | Provides historical context: locations repeatedly affected may show different expected human impacts.                                                                |
| Road / access exposure          | Roads, bridges and major access routes; optionally roads affected in historical events                                                     | Road Development Authority / NSDI; DMC for disaster damage                     | Shapefile/GeoJSON/GeoPackage; CSV for damage                        | Network/vector; map scale dependent                               | Optional feature for estimating isolation/access problems and potential displacement.                                                                                |
| Critical facilities             | Hospitals, schools, shelters and other critical facilities                                                                                 | Relevant government agencies / NSDI; OSM as supplementary source               | CSV/Excel + coordinates; GeoJSON/Shapefile                          | Point locations                                                   | Optional exposure/vulnerability feature; identifies populations near or dependent on critical facilities.                                                            |

| Factor                                   | Dataset Needed                                                                                    | Where to Request in Sri Lanka       | Format                 | Resolution                               | Purpose according to new component                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------|------------------------|------------------------------------------|----------------------------------------------------------------------------------------|
| Historical damage / displacement context | Houses fully/partially damaged, safe locations, families/persons evacuated, infrastructure damage | DMC situation reports / DesInventar | CSV/Excel; PDF reports | Event + District/DS level where reported | Improves prediction of displacement and total affected people and supports validation. |

**Recommended order of collection:** (1) DMC/DesInventar historical human-impact records, (2) population and age/vulnerability data from DCS, (3) flood and landslide severity outputs from Members 1 and 2, (4) rainfall and river-level data, and (5) GIS layers such as boundaries, DEM, flood extent and NBRO landslide hazard maps. The historical casualty/impact fields are the most important feasibility check before selecting the ML model.

**Source verification:** DMC's DesInventar describes a national disaster database with spatial/temporal event data and direct effects; DMC situation reports include affected people, deaths, injured, missing, housing damage and safe-location information. DCS provides GN-level population by sex and age and household information. NSDI provides District, DS and GN boundary layers. NBRO provides landslide hazard zonation at 1:50,000 and 1:10,000 scales. The Irrigation Department publishes river water-level and hydrometric station information.